Does a time crystal make a better quantum memory?

A disordered kicked-Ising (Floquet) spin chain used as a quantum reservoir computer, benchmarked against matched classical models on two real time series.

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1.32
memory capacity in the time-crystal regime (θ = 0.95π)

5.75
memory capacity in the scrambling regime (θ = 0.45π); ESN 5.31–7.82

AR
best forecaster on both series; ESN second, quantum reservoir third

1.3
scrambling-regime memory capacity left at 100 shots per readout (exact 5.17)

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

Does time-crystal order give a quantum reservoir better memory, and does any of it beat classical models on real data?

Quantum reservoir computing (Fujii & Nakajima 2017): let a disordered quantum system evolve, feed the input in, measure, and train only a linear readout. The physics does the feature engineering.

Discrete time crystals are driven spin chains that flip with period 2T and resist perturbations. Zhang et al. (2025) built a time-crystal reservoir and found phase boundaries best on synthetic tasks.

What we ask

- Which Floquet phase gives the most memory: time-crystal, scrambling, or weakly kicked?
- Does the best-memory phase also forecast real series best?
- Does it beat an echo-state network, a linear AR model and persistence with the **same features, splits and readout**?
- What would measurement cost on hardware?

Data (small, real, public; bundled with statsmodels)

Series	Values	Train	Val	Test
Sunspots (yearly)	309	200	47	62
Mauna Loa CO_2 (monthly change)	525	341	79	105

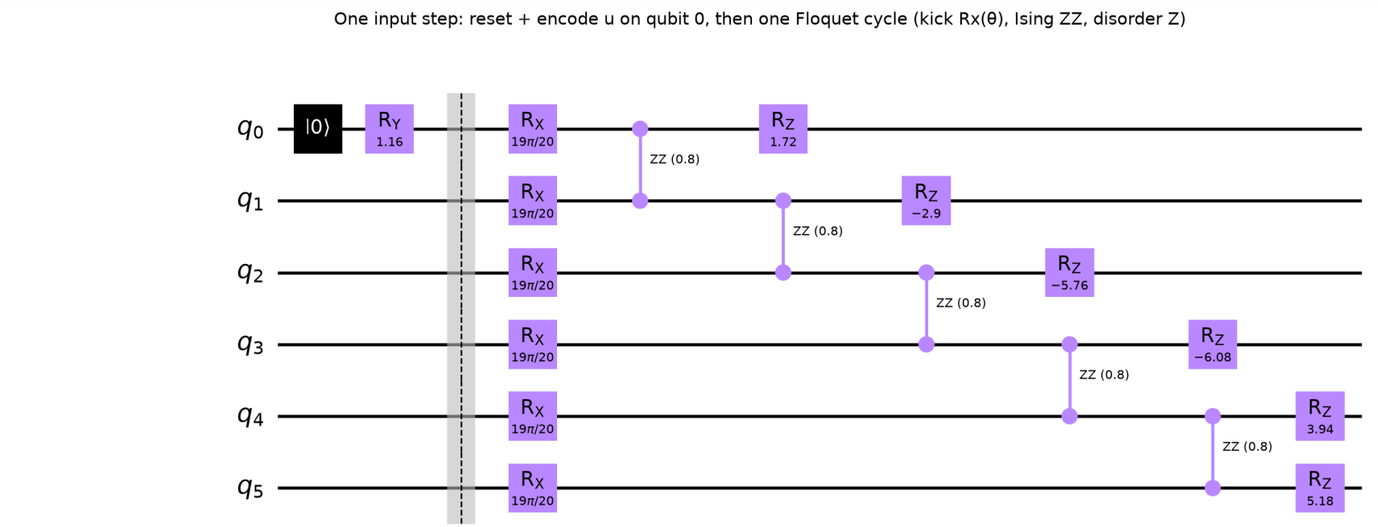
Yearly sunspot number 1700–2008 (309 values, statsmodels). Mauna Loa CO2, monthly change 1958–2001 (525 values, statsmodels). Time-ordered splits, train 65 % / validation 15 % / test 20 %.

Short answer

Time-crystal order **hurts** memory: it freezes the spins. The scrambling regime remembers as much as an 11-node ESN, but on real forecasting a linear AR model wins both series.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

A 6-spin kicked Ising chain, reset-and-inject input, 11 measured features



Circuit drawing: one step, one Floquet cycle, θ = 0.95π shown. The disorder angles h_i are drawn per seed.

1 · Inject
reset qubit 0 to $\sqrt{(1-u)}|0\rangle + \sqrt{u}|1\rangle$

2 · Evolve
2 Floquet cycles per step

3 · Read
$\langle Z_i \rangle$ and $\langle Z_i Z_{i+1} \rangle$

Floquet cycle $U_F = e^{-i \Sigma h_i Z_i} e^{-i J \Sigma Z_i Z_{i+1}} \Pi R_x(\theta)$, disorder $h_i \sim U[-\pi, \pi]$.

- **θ ≈ π, J ≠ 0:** time crystal (period-doubled magnetisation).
- **θ ≈ π/2:** scrambling (thermalising) kicks.
- **J = 0:** no interactions; information cannot leave qubit 0. Our quantum control.

Readout: 6 + 5 = 11 features → ridge regression. Only the readout is trained.

Simulated exactly with NumPy density matrices ($2^6 \times 2^6$). One reservoir step is cross-checked against Qiskit Aer's density-matrix simulator in `tests/`.

A fair benchmark: same features, same splits, same readout, tuned on validation only

Train 65 %
fit ridge readouts; min–max scaling fitted here only, not clipped

→

Validation 15 %
choose θ, ridge α, ESN ρ and input scale, AR order p

→

Refit
on train + validation with the chosen settings

→

Test 20 %
evaluated once; Diebold–Mariano tests

Models (one-step-ahead forecast, NMSE)

Model	Features	Tuned on validation
Quantum reservoir, J = 0.4	11	θ ∈ {0.2, 0.35, 0.5, 0.65, 0.8, 0.95} π, α
Quantum, J = 0 control	11	θ, α
Echo-state network	11 tanh nodes	ρ, input scale, α
Linear AR	p ≤ 11 lags	p, α
Persistence	–	–

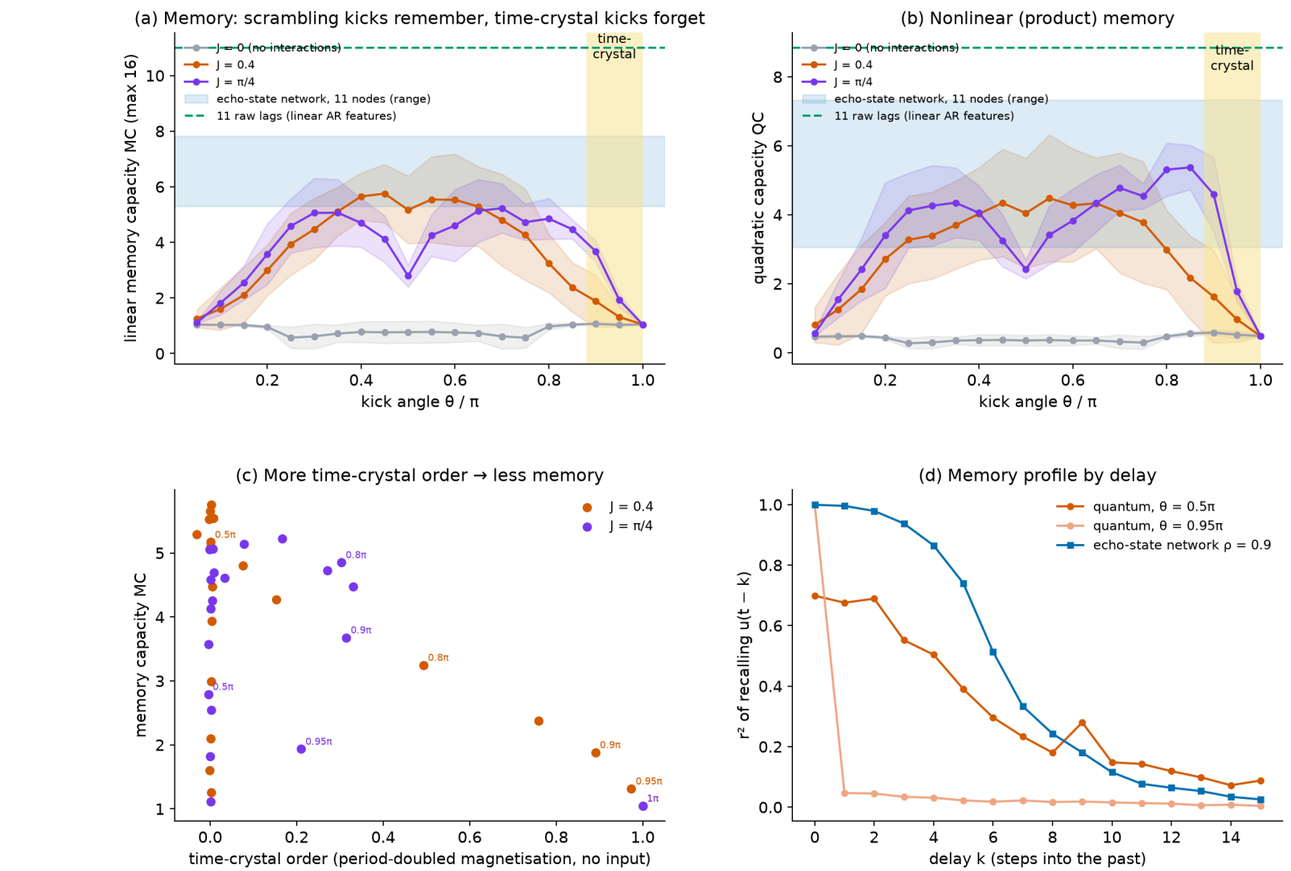
8 disorder / random seeds for every stochastic model. Memory capacity (E1): i.i.d. input, sum of r^2 for recalling u(t–k), k = 0…15.

Track 2 brief → what we did

- ✓ Real datasets, kept small (309 and 525 values).
- ✓ Same 11 features (or fewer) and the same split for every model.
- ✓ Min–max scaled on the training block only, not clipped (a test checks it).
- ✓ All hyperparameters chosen on validation; test touched once.
- ✓ Cross-validated: 8 seeds, plus 4-fold rolling-origin CV.
- ✓ Shows where the classical model wins, with Diebold–Mariano tests per seed.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

Time-crystal order hurts memory: it freezes the spins

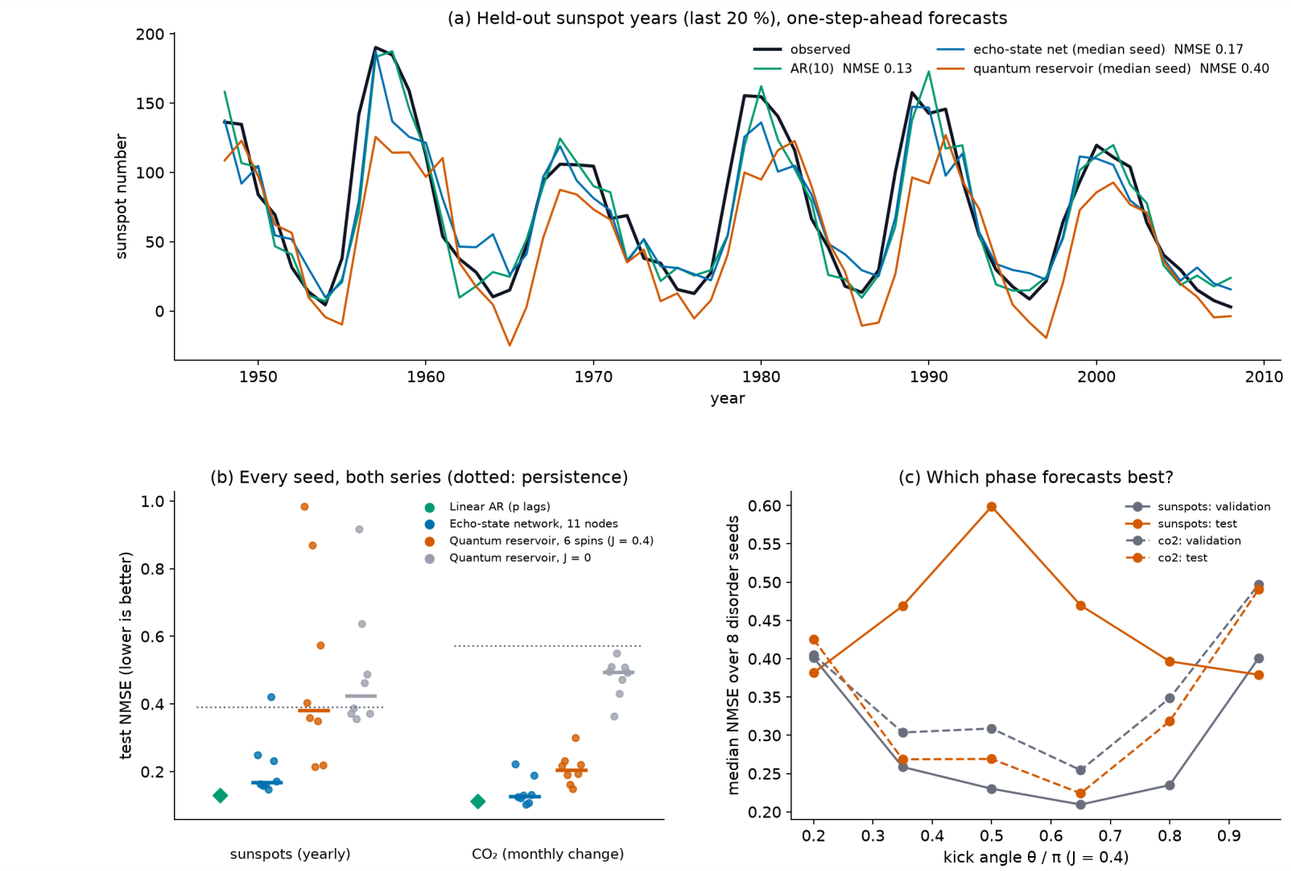


- **Scrambling kicks remember.** With J = 0.4, memory capacity is ≥ 5 for θ = 0.35π–0.65π, peaking at **5.75** (θ = 0.45π). That is inside the 11-node ESN range (5.31–7.82).
- **The time crystal forgets.** At θ = 0.95π the chain is strongly period-doubled (order 0.97) and memory capacity is only **1.32**: the spins mostly flip back and forth, and the input barely spreads.
- **Across the scan**, memory capacity anti-correlates with time-crystal order (r = -0.51, J ∈ {0.4, π/4}).
- **J = 0 control** stays at 0.57–1.07 at every θ: without interactions there is no reservoir.
- 11 raw lags would score 11.00: a linear delay line is the memory ceiling.

20 kick angles × 3 couplings × 5 disorder seeds; mean ± s.d. shaded.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

On real series, linear AR wins; the quantum reservoir is third



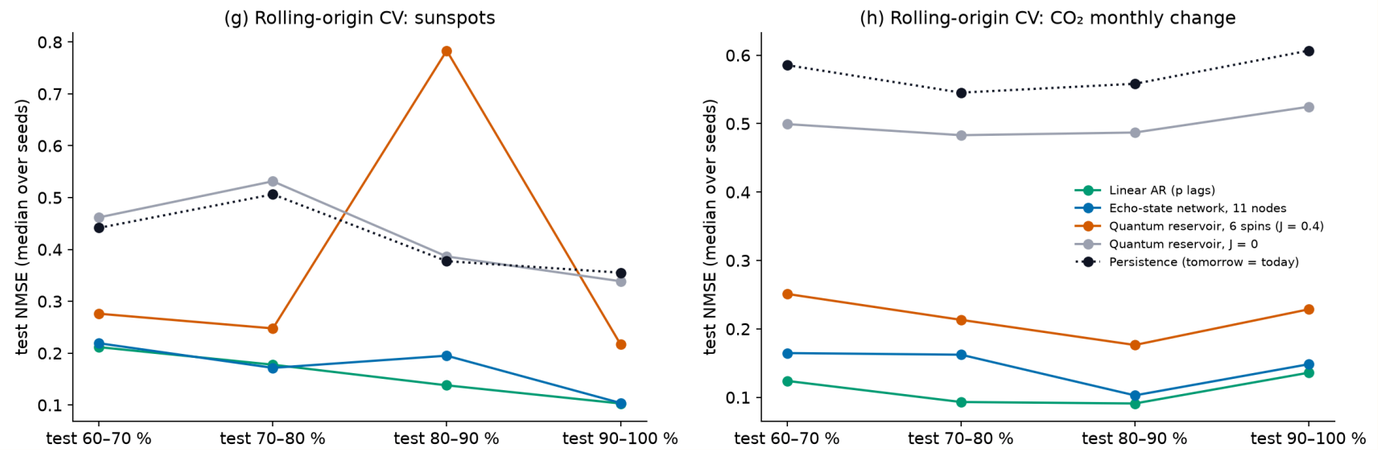
Test NMSE (median over 8 seeds)	Sunspots	CO_2
Linear AR (p lags)	0.131	0.112
Echo-state network, 11 tanh nodes	0.167 [0.15–0.42]	0.128 [0.10–0.22]
Quantum reservoir, 6 spins (J = 0.4)	0.381 [0.21–0.98]	0.205 [0.15–0.30]
Persistence (next = current)	0.391	0.572
Quantum reservoir, J = 0 (no interactions)	0.425 [0.36–0.92]	0.493 [0.36–0.55]

- **CO_2:** the quantum reservoir clearly beats its J = 0 control (0.21 vs 0.49) and persistence (0.57). Validation and test both prefer θ = 0.65π.
- **Sunspots:** on this split ≈ persistence (0.38 vs 0.39) and seed-sensitive (0.21–0.98). The phase that wins on validation (θ = 0.65π) does not win on test (θ = 0.95π).

[min–max over seeds]. AR order chosen on validation: p = 10 (sunspots), 11 (CO_2).

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

The ranking holds across 4 rolling-origin folds



Test NMSE over folds	sunspots median	sunspots mean rank	sunspots folds won	CO_2 median	CO_2 mean rank	CO_2 folds won
AR	0.16	**1.25**	3	0.11	**1**	4
ESN	0.18	**1.75**	1	0.16	**2**	0
Quantum reservoir	0.26	**3.5**	0	0.22	**3**	0
Persistence	0.41	**4**	0	0.57	**5**	0
J = 0 control	0.42	**4.5**	0	0.49	**4**	0

- sunspots, 4 folds: mean-rank order AR > ESN > Quantum reservoir > Persistence > J = 0 control; quantum reservoir mean rank 3.5 of 5, best in 0 of 4 folds.
- CO_2, 4 folds: mean-rank order AR > ESN > Quantum reservoir > J = 0 control > Persistence; quantum reservoir mean rank 3 of 5, best in 0 of 4 folds.

Rolling-origin cross-validation (4 folds) keeps the AR > ESN > quantum order on both series by mean rank. In those folds the quantum reservoir beats persistence on sunspots in median (0.26 vs 0.41) but is worse in 1 of 4 folds.

Each fold moves the test block forward and re-tunes every model on its own preceding validation block; per-fold numbers are medians over seeds.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

Classical wins everywhere we looked, and we say so

Diebold–Mariano tests on the test block (per seed, $p < 0.05$)

Series	AR significantly better than quantum	ESN significantly better than quantum	Quantum beats AR	ESN beats AR
Sunspots (yearly)	7 / 8	6 / 8	0 / 8	0 / 8
Mauna Loa CO_2 (monthly change)	7 / 8	5 / 8	0 / 8	2 / 8

Full ranking (median test NMSE)

Sunspots: AR 0.13 < ESN 0.17 < Quantum reservoir 0.38 < Persistence 0.39 < J = 0 control 0.42

CO_2: AR 0.11 < ESN 0.13 < Quantum reservoir 0.21 < J = 0 control 0.49 < Persistence 0.57

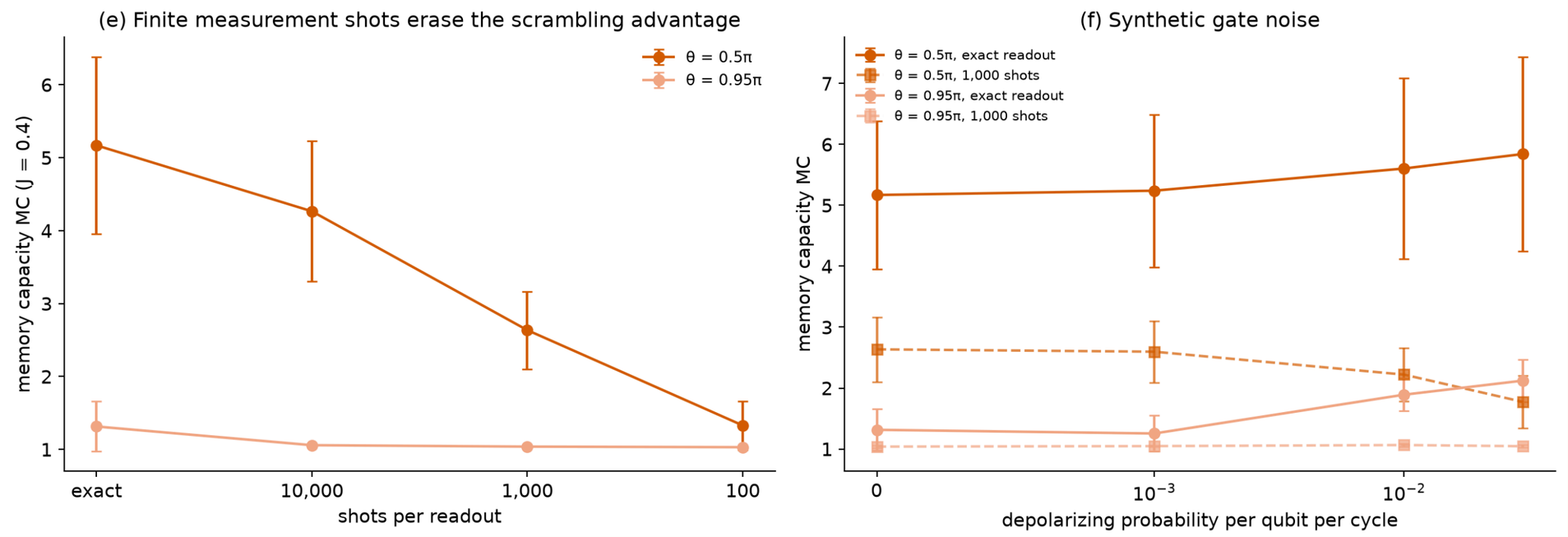
Why classical wins here

- **Both series are close to linear-autoregressive.** Sunspots have an ~11-year cycle; CO_2 changes have a 12-month cycle. AR with p ≈ 10–11 lags captures exactly that, and our reservoir remembers only ~6 steps.
- **An 11-node ESN has the same feature count**, more memory than the best quantum phase, and costs microseconds.
- **Reservoir choice is fragile on sunspots:** θ picked on validation varies by seed (0.5π ×2, 0.65π ×4, 0.8π ×2).

> **What the quantum reservoir does show:** interactions matter. On CO_2 the J = 0 control is 2.4× worse, so the forecasting skill comes from many-body dynamics, not the single-qubit encoding.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

Finite shots erase the scrambling advantage; hardware would pay a restart bill



Memory capacity	θ = 0.5π	θ = 0.95π
exact	5.17	1.32
10,000 shots	4.27	1.06
1,000 shots	2.64	1.04
100 shots	1.33	1.03

Depolarizing p	exact	1,000 shots
0	5.17	2.64
0.001	5.24	2.60
0.01	5.60	2.22
0.03	5.84	1.77

θ = 0.5π, J = 0.4; mean over seeds.

The time crystal is shot-robust only because it has almost no memory to lose.

Hardware: a projective readout destroys the state, so the sequence is re-run up to every readout time: 95,790 Floquet cycles per shot for the 309-year sunspot series instead of 618 (155×), and 309,000 circuit runs at 1,000 shots. Our whole E2 protocol would need ≈ 40 million runs. Windowed replay or weak measurement (Mujal et al. 2023) would cut this.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

An independent replication and extension, not the first time-crystal reservoir

Work	Reservoir	Finding
Fujii & Nakajima 2017	disordered spins	QRC; reset-and-inject input
Martínez-Peña et al. PRL 2021	static disordered Ising	thermal phase / transition best; localisation hurts
Mujal et al. 2023	measurement protocols	restart / weak-measurement cost set by memory time
Zhang et al. 2025 (arXiv:2508.15230)	**DTC kicked Ising**, same input and readout	phase boundaries best on memory, parity, NARMA10; MNIST on hardware

Zhang et al. is the closest work and essentially our model. Our scan was done independently; see `docs/literature.md`.

What this project adds

- **Real series:** sunspots and CO_2, not only synthetic tasks.
- **Matched classical baselines:** an ESN with the same 11 features, linear AR, persistence, and a J = 0 quantum control, all tuned on validation only.
- **Memory phase ≠ forecasting phase:** on sunspots the validation-best θ does not hold on test.
- **Measurement cost:** shot noise and restart bill per phase.

> Consistent with prior work: deep in the time-crystal phase the reservoir is a poor memory; the useful regime is the scrambling side, toward the boundary.

Classical simulation — no quantum hardware used. NumPy density matrices; reservoir step checked against Qiskit Aer.

Time-crystal order is robust because it does not listen

- **Memory:** scrambling kicks reach ESN-level memory (5.75); the time crystal stays at 1.32.
- **Forecasting:** linear AR wins both real series; the ESN is second; the quantum reservoir is third and beats its J = 0 control.
- **Hardware:** with 1,000 shots the scrambling advantage shrinks to 2.64, and every readout needs a restart.
- **No quantum advantage claimed.** The useful output is a fair, reusable benchmark for the next reservoir idea.

Reproduce (≈ 90 s on a laptop)

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.lock.txt && pip install -e .
python -m qreservoir.experiments
python -m qreservoir.figures
pytest
streamlit run app/app.py
python presentation/build.py
```

github.com/wdreisz/floquet-reservoir · MIT licence

Every number in this deck is generated from `artifacts/*.json` by `presentation/build.py`. Python 3.12.13, Qiskit 2.6.0, NumPy 2.5.4.